Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce Department of Information Technology (B.Sc.I.T Semester V) Dot .Net Core and Progressive Web Application Practical

Practical – VII

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Class: TYIT	Batch: 01
Date of Assignment: 18/08/2026	Date/Time of Submission: 18/08/2026

a. Develop CRUD operations using ASP.NET Core Web API and Entity Framework Core.

Code :

Program.cs:-

```
using Microsoft.EntityFrameworkCore;
using StudentAPIWithEF.Data;

var builder = WebApplication.CreateBuilder(args);
builder.Services.AddControllers();

builder.Services.AddDbContext<AppDbContext>(options => options.UseSqlServer(
   builder.Configuration.GetConnectionString("DefaultConnection")));

builder.Services.AddEndpointsApiExplorer();
builder.Services.AddSwaggerGen();

var app = builder.Build();

app.UseSwagger();
app.UseSwaggerUI();
```

```
app.UseHttpsRedirection();

app.UseAuthorization();

app.MapControllers();


app.Run();
```

StudentController.cs:-

```
using Microsoft.AspNetCore.Mvc;

using StudentAPIWithEF.Data;

using StudentAPIWithEF.Models;


namespace StudentAPIWithEF.Controllers

{

  [Route("api/[controller]")]

  [ApiController]

  public class StudentsController : ControllerBase

  {

    private readonly AppDbContext _context;


    public StudentsController(AppDbContext context)

    {

      _context = context;

    }


    // GET: api/Students

    [HttpGet]

    public IActionResult GetStudents()

    {

      return Ok(_context.Students.ToList());

    }


    // GET: api/Students/1
```

```
[HttpGet("{id}")]
public IActionResult GetStudent(int id)
{
    var student = _context.Students.Find(id);

    if (student == null)
        return NotFound();

    return Ok(student);
}

// POST: api/Students
[HttpPost]
public IActionResult AddStudent(Student student)
{
    _context.Students.Add(student);
    _context.SaveChanges();

    return Ok(student);
}

// PUT: api/Students/1
[HttpPut("{id}")]
public IActionResult UpdateStudent(int id, Student student)
{
    var data = _context.Students.Find(id);

    if (data == null)
        return NotFound();

    data.Name = student.Name;
    data.Age = student.Age;
    data.Course = student.Course;
```

```
            _context.SaveChanges();

            return Ok(data);
        }

        // DELETE: api/Students/1
        [HttpDelete("{id}")]
        public IActionResult DeleteStudent(int id)
        {
            var student = _context.Students.Find(id);

            if (student == null)
                return NotFound();

            _context.Students.Remove(student);
            _context.SaveChanges();

            return Ok();
        }
    }
}
```

Appsettings.json:-

```
{
  "Logging": {
    "LogLevel": {
      "Default": "Information",
      "Microsoft.AspNetCore": "Warning"
    }
  },
  "AllowedHosts": "*",
```

```json
  "ConnectionStrings": {
    "DefaultConnection": "Server=(localdb)\\MSSQLLocalDB;Database=StudentDB;Trusted_Connection=True;"
  }
}
```

AppDbContext.cs:-

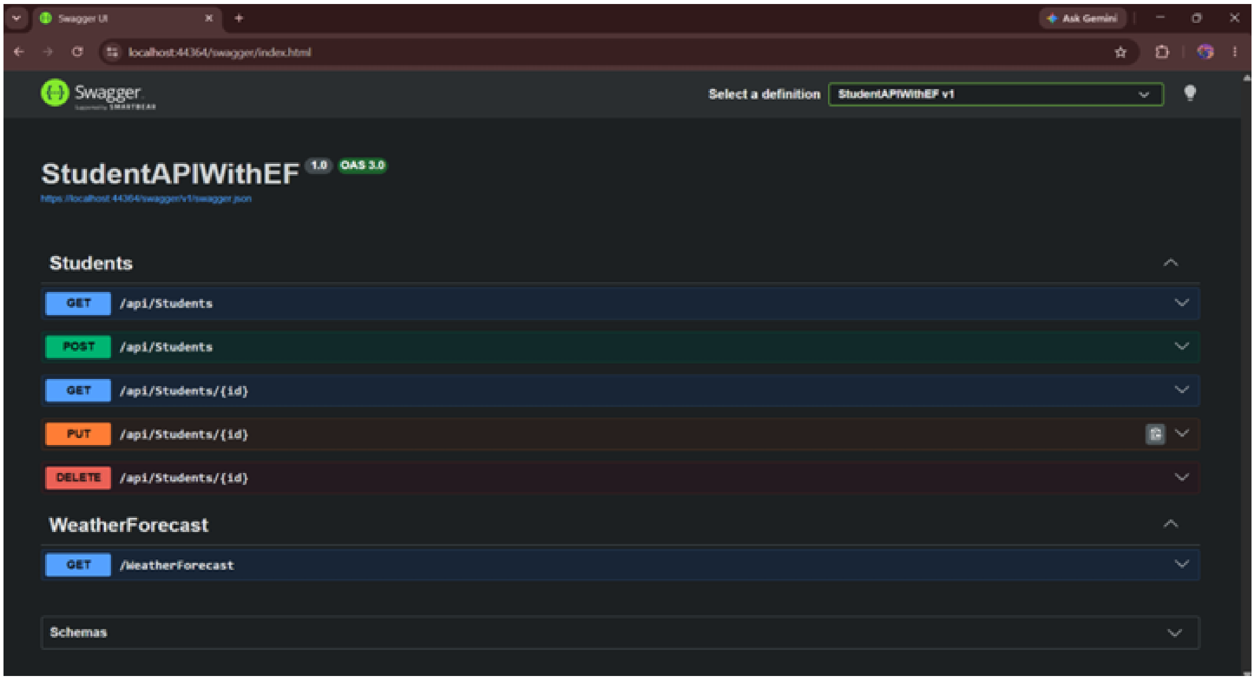
```csharp
using Microsoft.EntityFrameworkCore;
using StudentAPIWithEF.Models;
namespace StudentAPIWithEF.Data
{
    public class AppDbContext:DbContext
    {
        public AppDbContext(DbContextOptions<AppDbContext> options) : base(options)
        {
        }
        public DbSet<Student> Students { get; set; }
    }
}
```

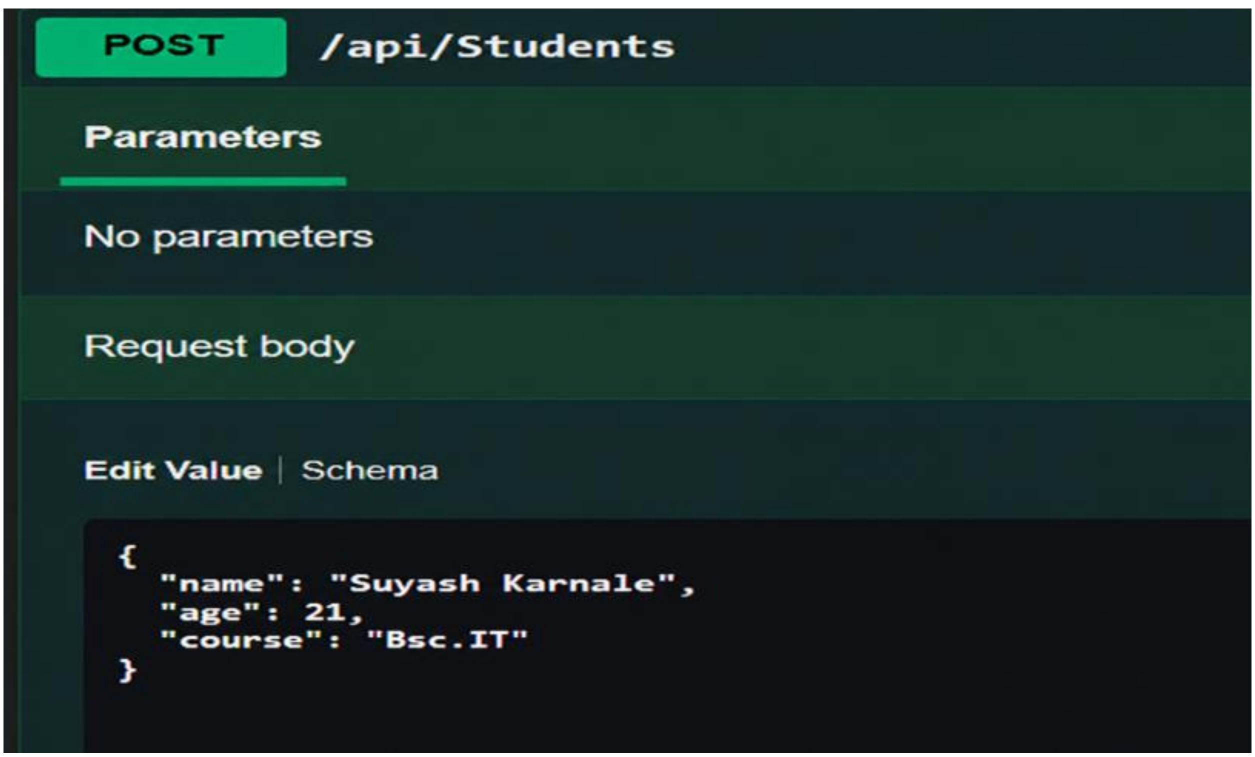
Student.cs:-

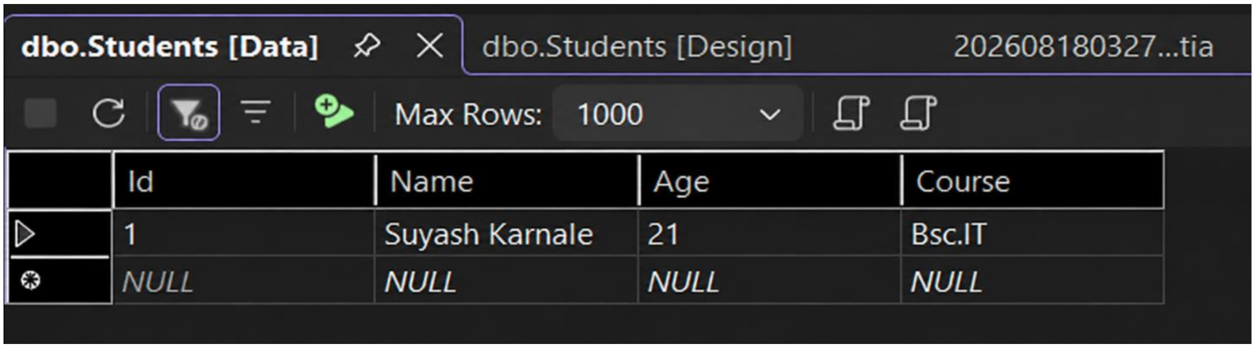
```csharp
namespace StudentAPIWithEF.Models
{
    public class Student
    {
        public int ID { get; set; }
        public required string Name { get; set; }
        public int Age { get; set; }
        public required string Course { get; set; }

    }
}
```

Output:-


Swagger UI
localhost:44364/swagger/index.html
Swagger
Select a definition
StudentAPIWithEF v1
StudentAPIWithEF 1.0 OAS 3.0
Students
GET /api/Students
POST /api/Students
GET /api/Students/{id}
PUT /api/Students/{id}
DELETE /api/Students/{id}
WeatherForecast
GET /WeatherForecast
Schemas


POST /api/Students
Parameters
No parameters
Request body
Edit Value | Schema
{
"name": "Suyash Karnale",
"age": 21,
"course": "Bsc.IT"
}



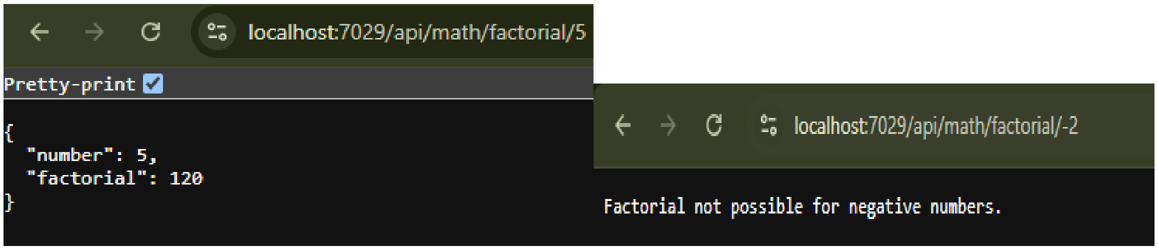
	Id	Name	Age	Course
	1	Suyash Karnale	21	Bsc.IT
	NULL	NULL	NULL	NULL

b. To create an ASP.NET Core Web API endpoint that calculates the factorial of a given number and test the API using Postman.

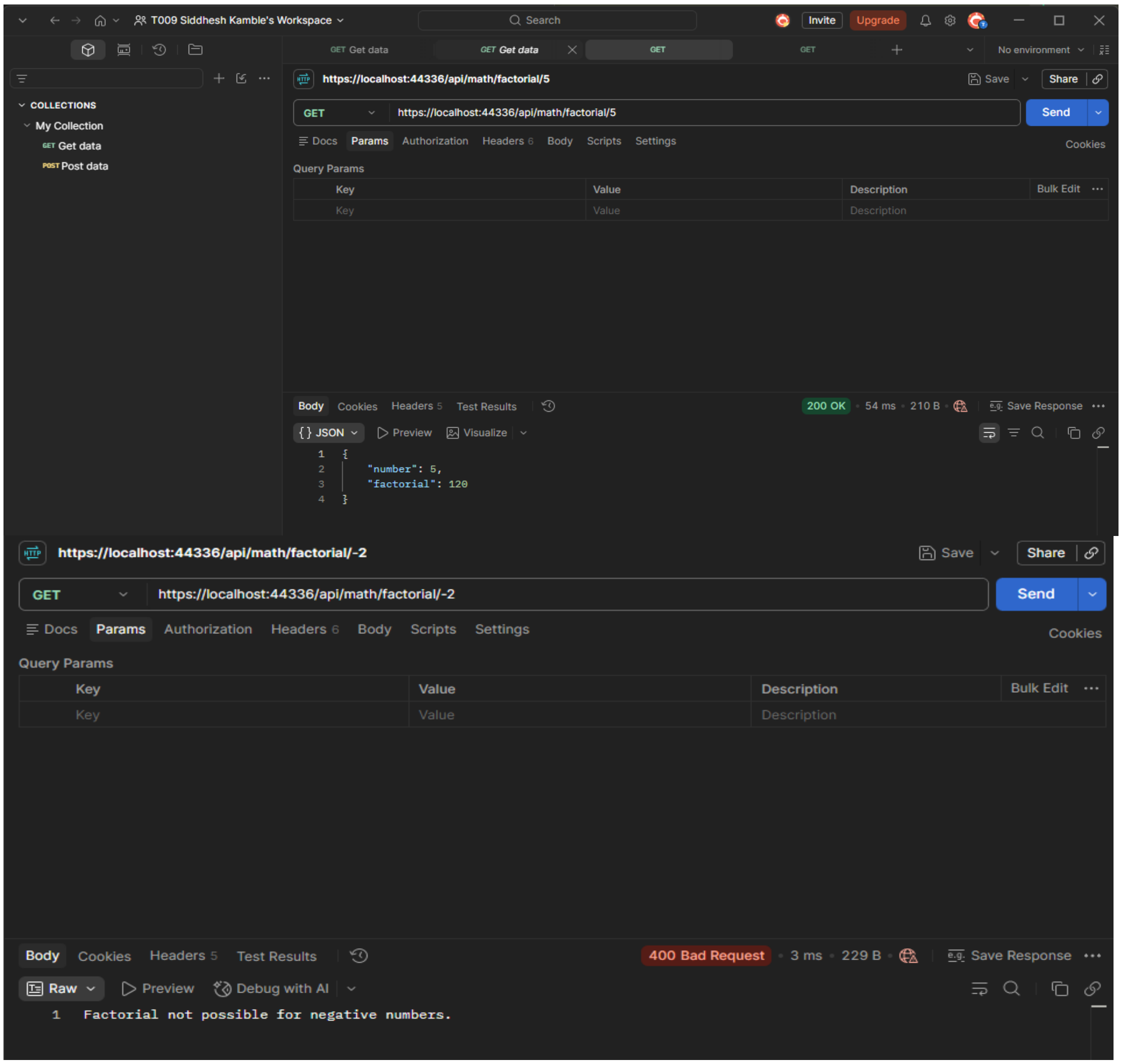
Code :

```
using Microsoft.AspNetCore.Mvc;
namespace FactorialAPI.Controllers
{
  [Route("api/[controller]")]
  [ApiController]
  public class MathController : ControllerBase
  {
    [HttpGet("factorial/{number}")]
    public IActionResult GetFactorial(int number)
    {
      if (number < 0)
      {
        return BadRequest("Factorial not possible for negative numbers.");
      }
      long factorial = 1;
      for (int i = 1; i <= number; i++)
      {
        factorial *= i;
      }
      return Ok(new
      {
        Number = number,
        Factorial = factorial
      });
    }
  }
}
```

Output :


localhost:7029/api/math/factorial/5
Pretty-print
{
"number": 5,
"factorial": 120
}
localhost:7029/api/math/factorial/-2
Factorial not possible for negative numbers.

Postman :


T009 Siddhesh Kamble's Workspace
Search
Invite
Upgrade
No environment
COLLECTIONS
My Collection
GET Get data
POST Post data
https://localhost:44336/api/math/factorial/5
GET
https://localhost:44336/api/math/factorial/5
Send
Save
Share
Docs Params Authorization Headers 6 Body Scripts Settings
Cookies
Query Params
Key Value Description Bulk Edit
Body Cookies Headers 5 Test Results
200 OK 54 ms 210 B
Save Response
{} JSON Preview Visualize
{
"number": 5,
"factorial": 120
}
https://localhost:44336/api/math/factorial/-2
GET
https://localhost:44336/api/math/factorial/-2
Send
Docs Params Authorization Headers 6 Body Scripts Settings
Query Params
Key Value Description Bulk Edit
Body Cookies Headers 5 Test Results
400 Bad Request 3 ms 229 B
Save Response
Raw Preview Debug with AI
Factorial not possible for negative numbers.